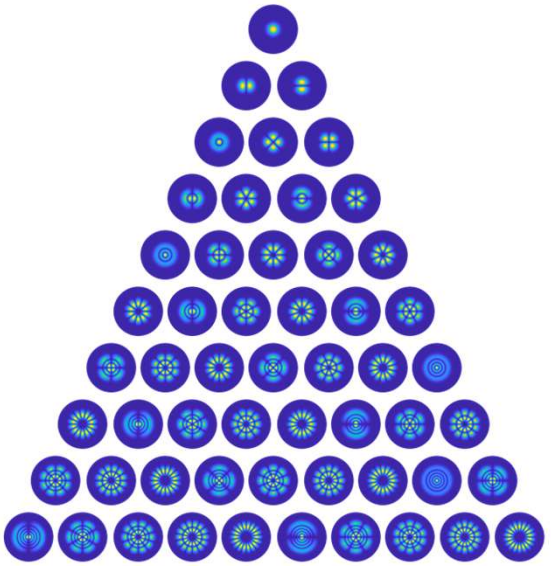
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

Lecture 5



Prof. Dr.-Ing. Georg Rademacher

University of Stuttgart
Faculty 5: Computer Science, Electrical Engineering and Information Technology



25.-27. November 2025

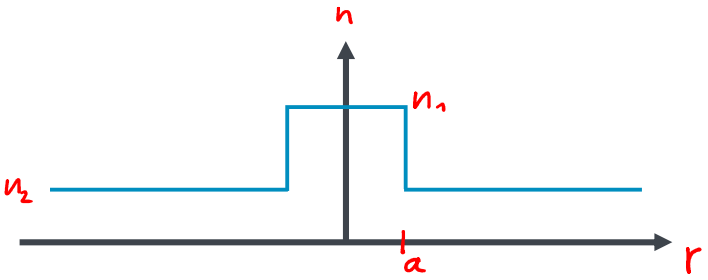
#InternationalDays Fakultät 5



- ✓ Entdecke Möglichkeiten für dein Auslandssemester!
- ✓ Lerne Partneruniversitäten kennen und spreche mit Austauschstudierenden!
- ✓ Feier mit uns die kulturelle Vielfalt unserer Fakultät - mit einem Programm von Studierenden für Studierende!

ISP International Service Point

Linearly polarized (LP) waves



$$n(r) = \begin{cases} n_1 & r \le a \\ n_2 & a < r \le \frac{D}{2} \end{cases}$$

$$a^2 \nabla_t^2 \psi_1 + u^2 \psi_1 = 0 \qquad r \le a$$

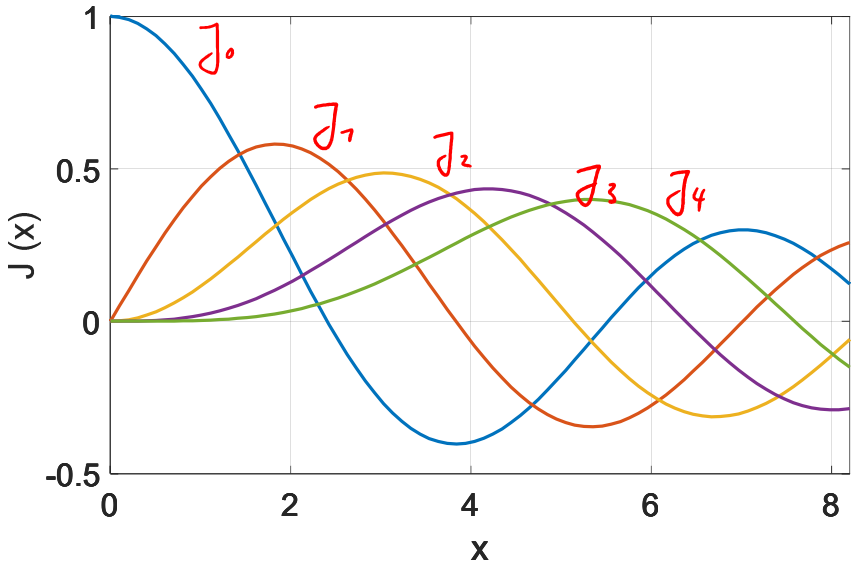
$$a^2 \nabla_t^2 \psi_2 + v^2 \psi_2 = 0 \qquad r \ge a$$

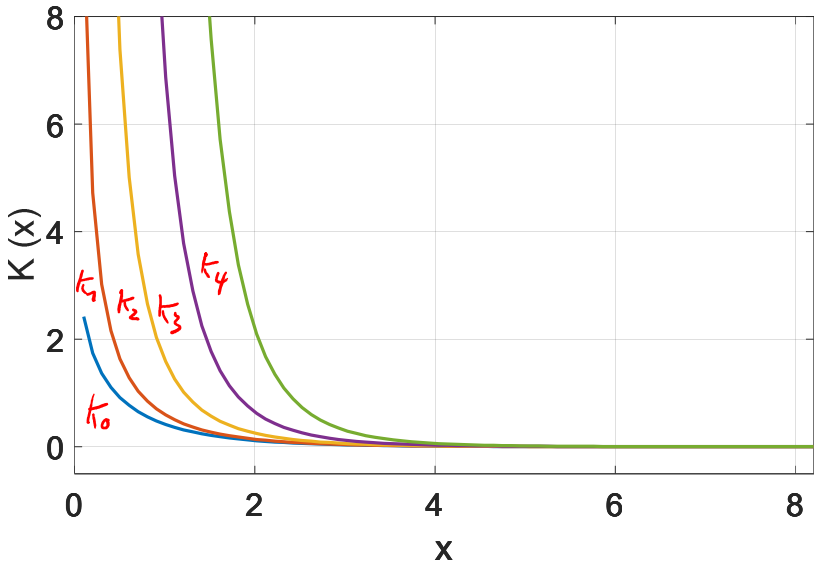
$$\psi_1(r,\varphi) = A_1 J_l\left(r \frac{u}{a}\right) \begin{Bmatrix} \cos(l\cdot\varphi) \\ \sin(l\cdot\varphi) \end{Bmatrix}$$

$$\psi_2(r,\varphi) = A_2 K_l\left(r \frac{v}{a}\right) \begin{Bmatrix} \cos(l\cdot\varphi) \\ \sin(l\cdot\varphi) \end{Bmatrix}$$

Besselfunctions

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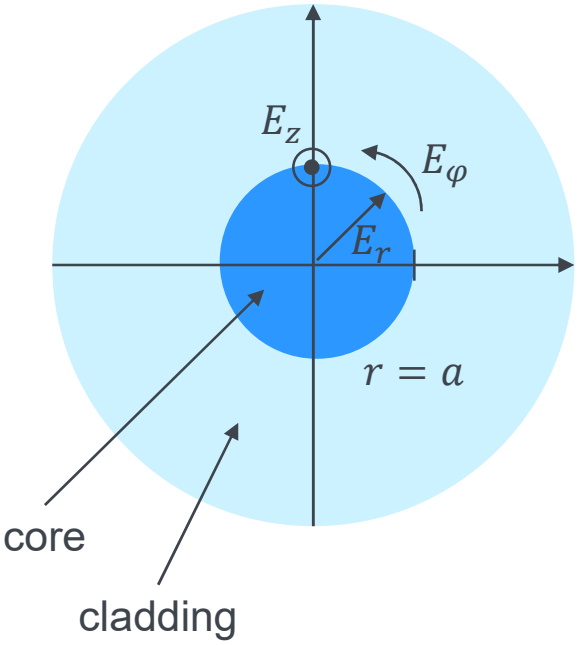




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Bonus Test

Which field component doesn't need to be continuous at r=a?



A E_z

B E_φ

C E_r

https://ilias3.uni-stuttgart.de/goto.php?target=xlvo_1_pin_UVJY

Linearly polarized (LP) waves

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$$\psi_1\big|_{r=a} = \psi_2\big|_{r=a} \quad \Rightarrow \quad A_1 J_\ell(u) = A_2 K_\ell(v)$$

$$\frac{\partial \psi_1}{\partial r}\Big|_{r=a} = \frac{\partial \psi_2}{\partial r}\Big|_{r=a} \quad \Rightarrow \quad A_1 \frac{u}{a} J_\ell'(u) = A_2 \frac{v}{a} K_\ell'(v)$$

$$\frac{u \cdot J_\ell'(u)}{J_\ell(u)} = \frac{v \cdot K_\ell'(v)}{K_\ell(v)}$$

Linearly polarized (LP) waves

$$J_l'(x) = -J_{l+1}(x) + \frac{l}{x} J_l(x)$$

$$K_l'(x) = -K_{l+1}(x) + \frac{l}{x} K_l(x)$$

$$-\frac{u\, J_{l+1}(u)}{J_l(u)} + \frac{v \cdot K_{l+1}(v)}{K_l(v)} = 0 = f(u)$$

Linearly polarized (LP) waves

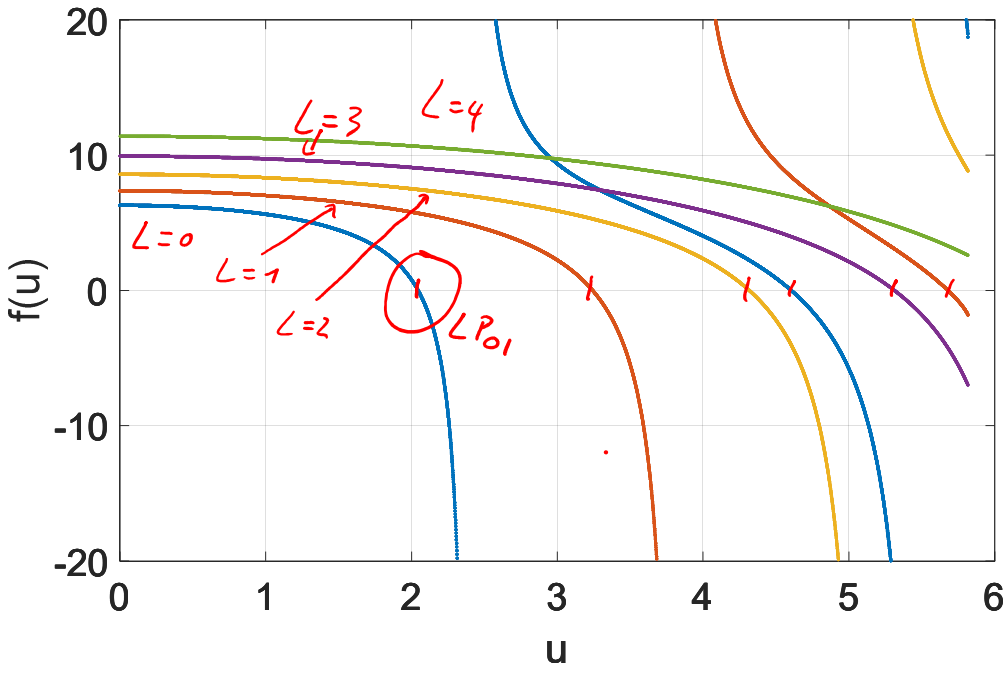
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$\lambda = 1{,}55\,\mu m,\ a = 7\,\mu m,\ NA = 0.205 \rightarrow V = 5.817$

$V = k_0 \cdot a \cdot NA$

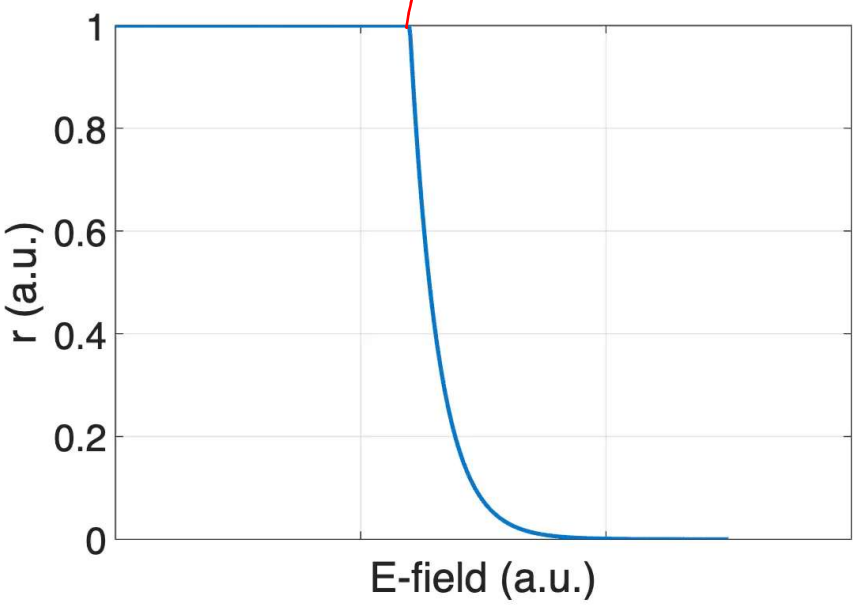
$V^2 = u^2 + v^2$

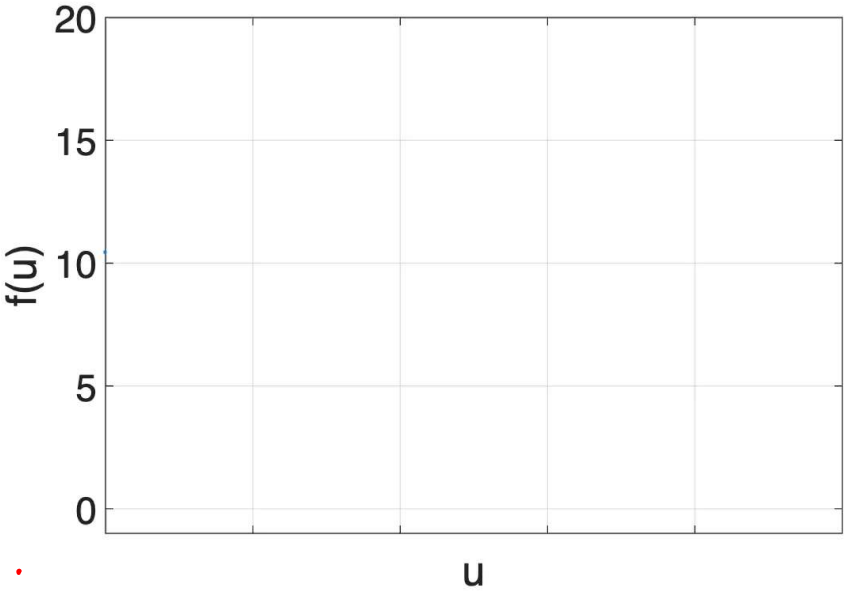
$\rightarrow v = \sqrt{V^2 - u^2}$



Linearly polarized (LP) waves

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Linearly polarized (LP) waves



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for given V countable solutions of circumferential order L and radial order P

LP_{LP} - Modes



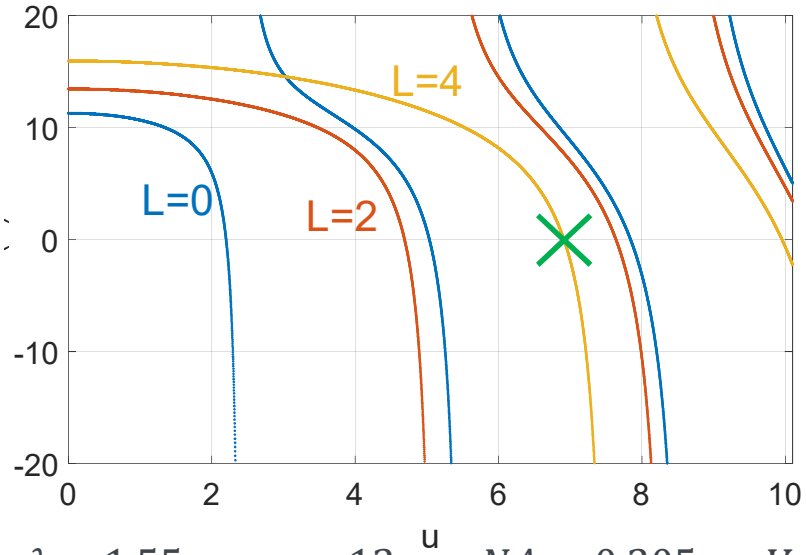
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Concept Test

Concept test

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Germany

What is the mode number of LP mode that corresponds to the marked zero?



A LP42

B LP41

C LP21

$\lambda = 1.55\ \mu m, a = 12\ \mu m, NA = 0.205\ \rightarrow V = 9.97$

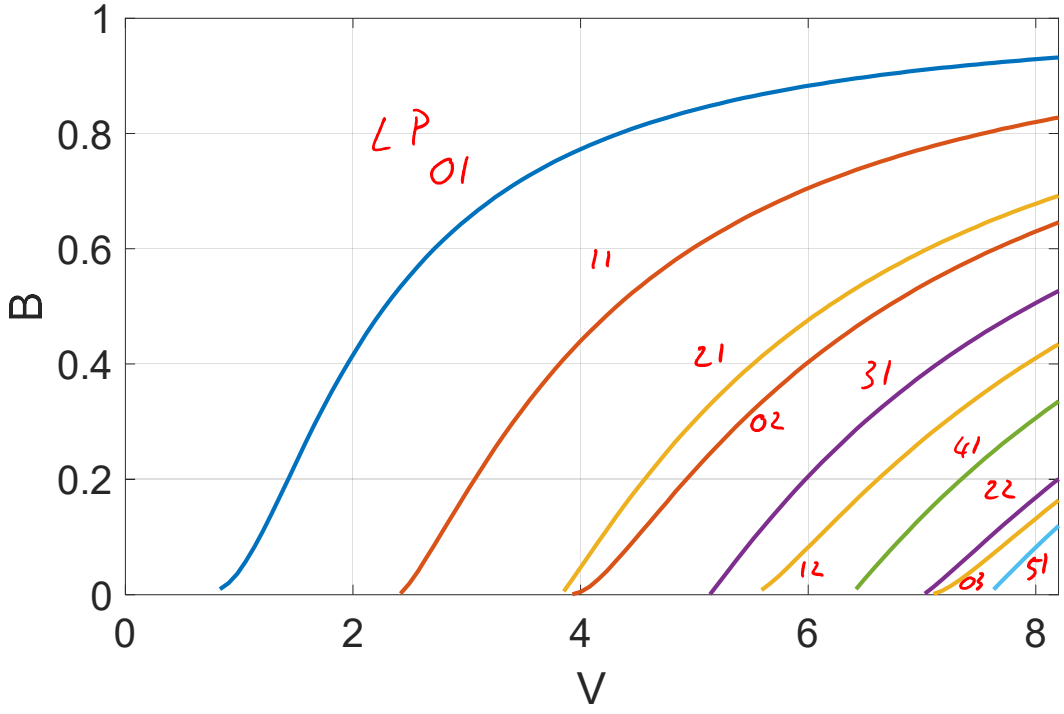
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Linearly polarized (LP) waves

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$$V = k_0 \cdot a \cdot NA$$
$$= k_0 \cdot a \cdot \sqrt{n_1^2 - n_2^2}$$

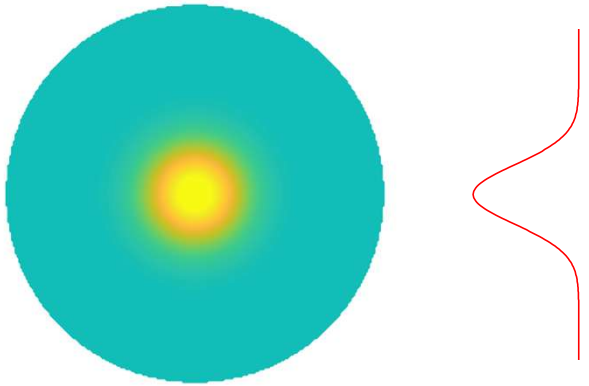
$$B \approx \frac{\frac{\beta}{k_0} - n_2}{n_1 - n_2}$$

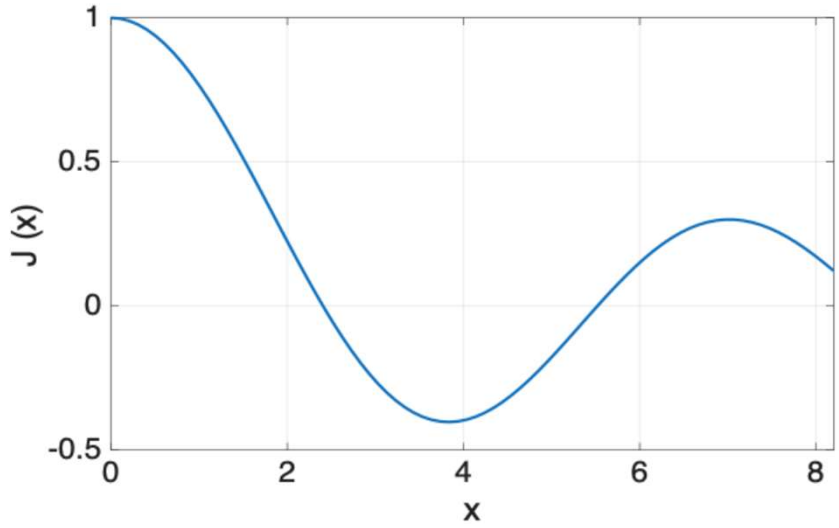


Mode Fields in Step Index Fibers

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LP_{01}



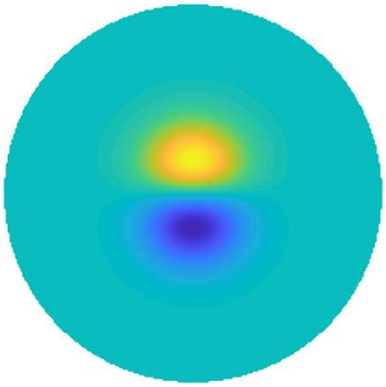


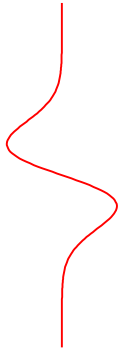
$$\psi_{lm}(r,\varphi) = A_{lm} J_l\left(r\,\frac{u}{a}\right) \begin{cases} \cos(l\cdot\varphi) \\ \sin(l\cdot\varphi) \end{cases}$$

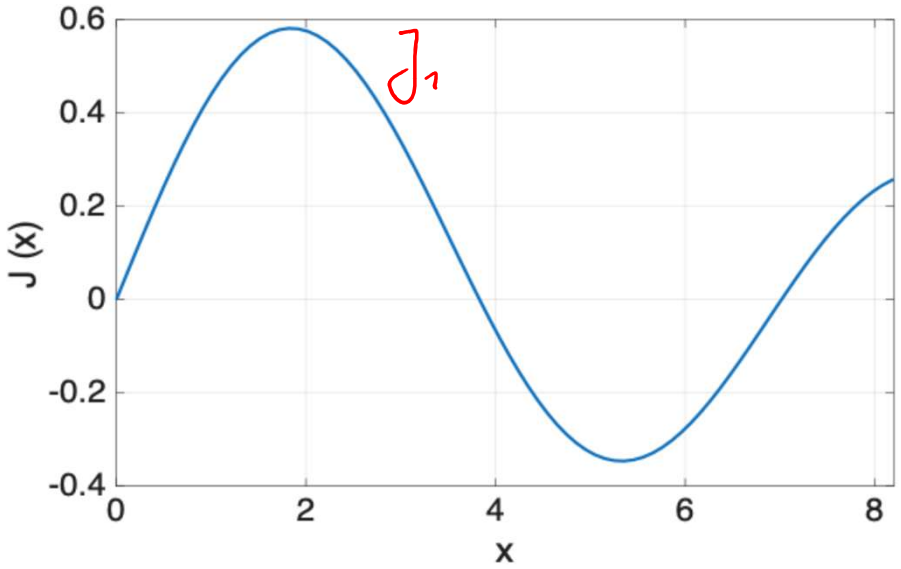
Mode Fields in Step Index Fibers

University of Stuttgart
Germany

$LP_{1,b}$

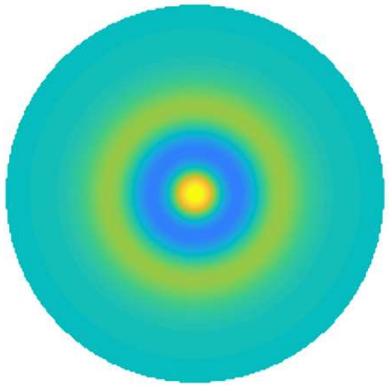


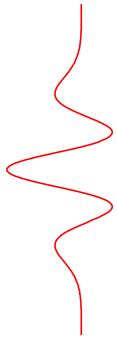


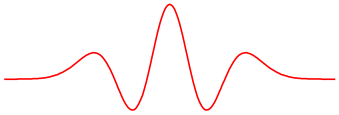


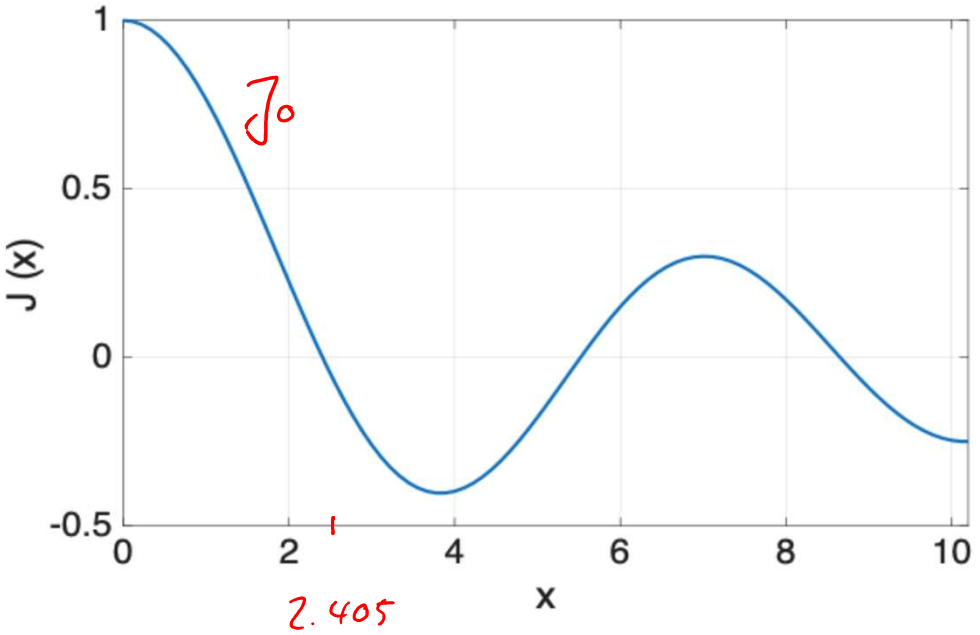
$$\psi_{1}(r,\varphi) \sim J_1\left(r\tfrac{u}{a}\right)\begin{cases}\cos(1\cdot\varphi) & a\\ \sin(1\cdot\varphi) & b\end{cases}$$

Mode Fields in Step Index Fibers

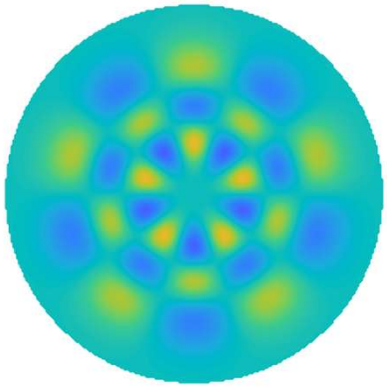




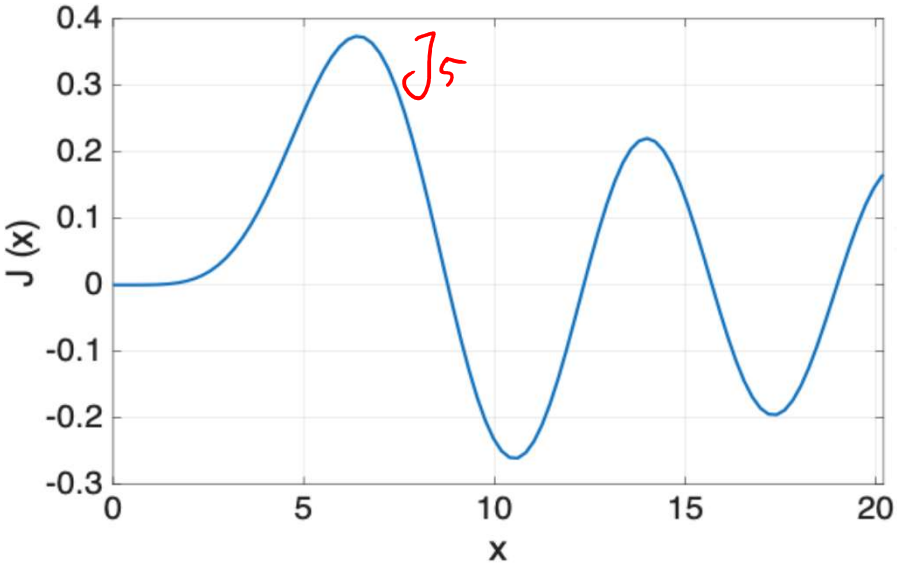




Mode Fields in Step Index Fibers







Single-mode fiber

only supports LP_{01}

$$\frac{u\,J_1(u)}{J_0(u)} - \frac{v\,K_1(u)}{K_0(v)} = 0$$

$1.5 < V < 2.5$

$$v = 1.1428 \cdot V - 0.996$$

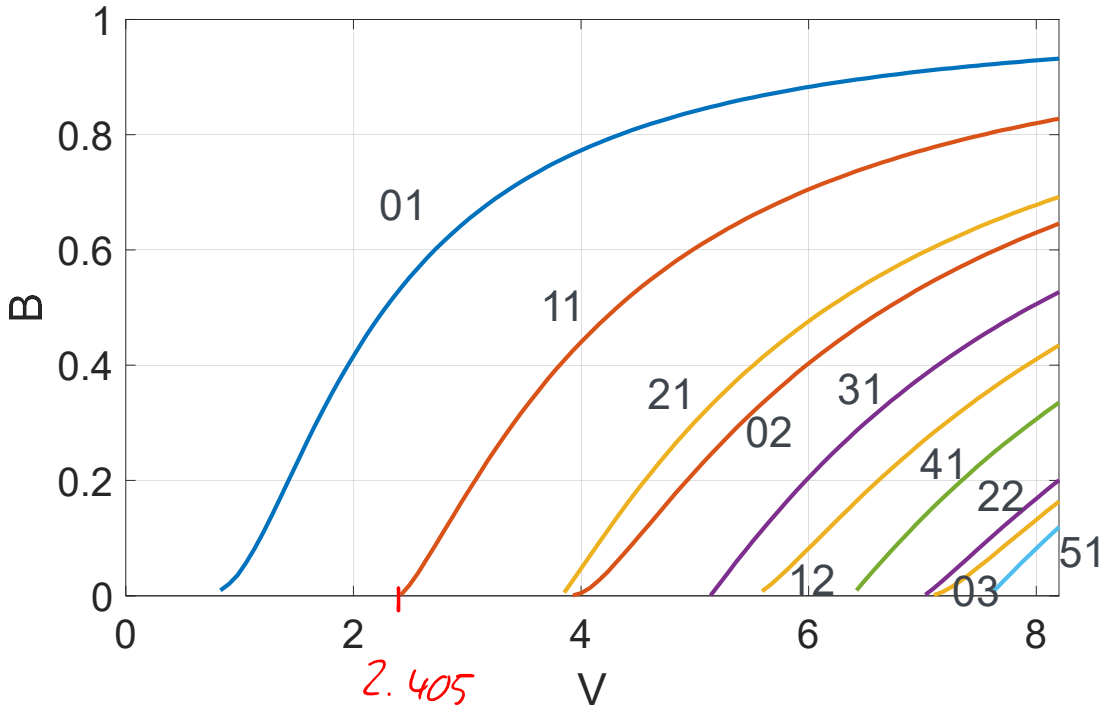
Single-mode fiber

$V = 0 \Rightarrow LP_{11}$ mode starts propagating

$$\frac{u_c J_0(u_c)}{J_1(u_c)} = 0 \qquad \Rightarrow J_0(u_c) = 0$$

$$V_c = 2.405$$

Linearly polarized (LP) waves



Chromatic dispersion

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Germany

$$\tau = \frac{d\beta}{d\omega}$$

$$\beta = k_0 \left(B(n_1 - n_2) + n_2 \right)$$

$$\tau = \frac{d\left(k_0 \left(B(n_1 - n_2) + n_2 \right)\right)}{d\omega}$$

$$\frac{dn_1}{d\omega} = \frac{dn_2}{d\omega}$$

$$\Rightarrow \frac{d(n_1 - n_2)}{d\omega} = 0$$

$$\Rightarrow \frac{d\,NA}{d\omega} = \frac{d\sqrt{n_1^2 - n_2^2}}{d\omega} \approx 0$$

Chromatic dispersion

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Germany

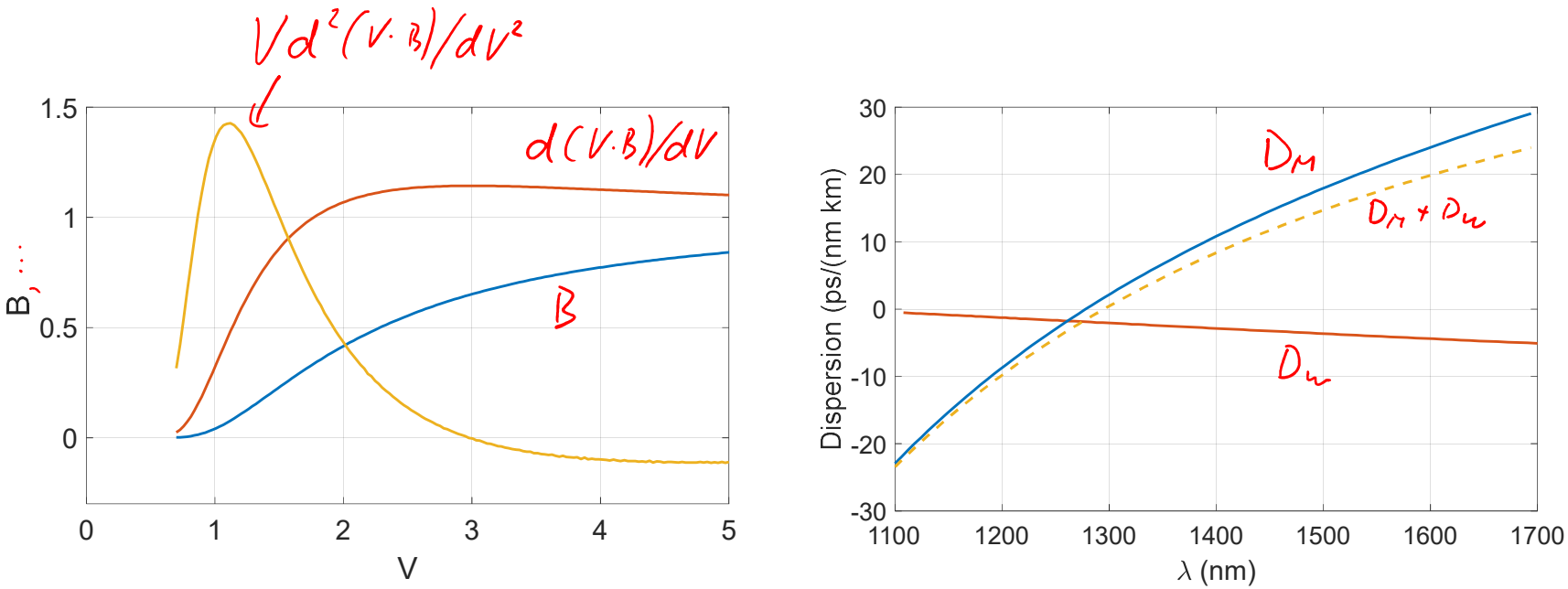
$$\tau = (n_1 - n_2)\frac{d(k_0 B)}{d\omega} + \frac{d(k_0 n_2)}{d\omega} = \frac{n_1 - n_2}{c} \cdot \frac{d(V \cdot B)}{dV} + \frac{1}{c} N_2$$

$$\frac{d\tau}{d\lambda} = \underbrace{-\frac{n_1 - n_2}{c \cdot \lambda} \cdot \frac{V d^2(V \cdot B)}{dV^2}}_{D_W} \underbrace{- \frac{\lambda}{c} \cdot \frac{d^2 n_2}{d\lambda^2}}_{D_M}$$

D_W : Waveguide dispersion

D_M : Material dispersion

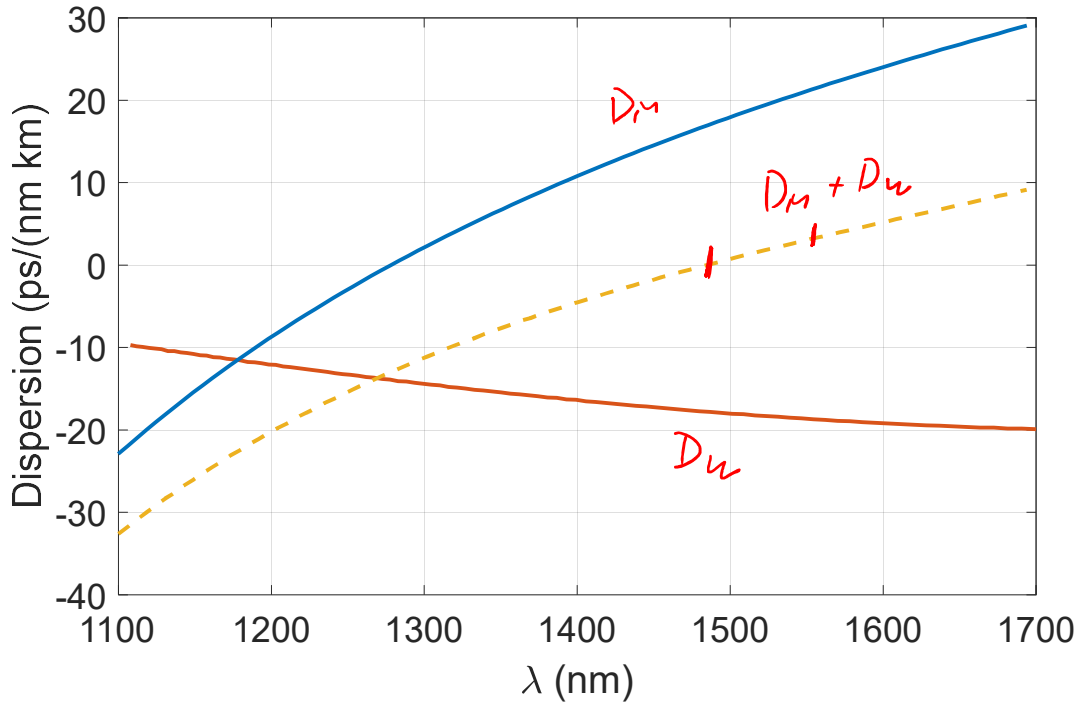
Chromatic dispersion of the fundamental mode



Chromatic dispersion of the fundamental mode

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$\lambda = 1.1 - 1.7\,\mu m$, $a = 2.4\,\mu m$, $NA = 0.15$





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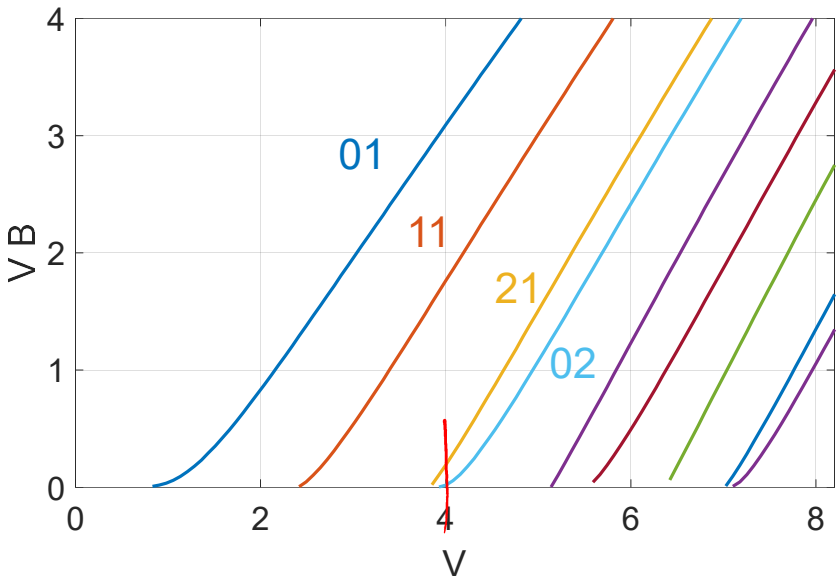
Concept Test

Concept test

$$\tau = \frac{n_1 - n_2}{c} \cdot \frac{d(V \cdot B)}{dV} + \frac{1}{c} N_2$$

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Which mode has the smallest group delay at V = 4?



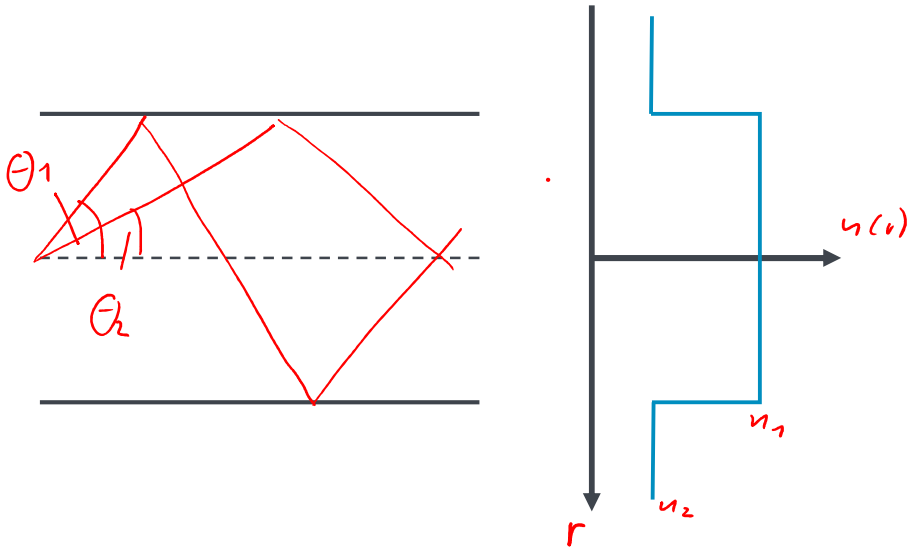
A LP01
B LP21
C LP02

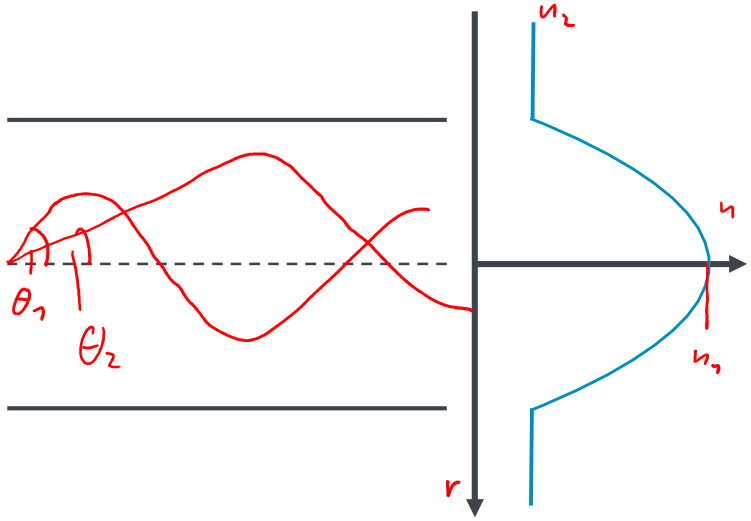
https://ilias3.uni-stuttgart.de/vote/**WXHM**

Graded-index fibers

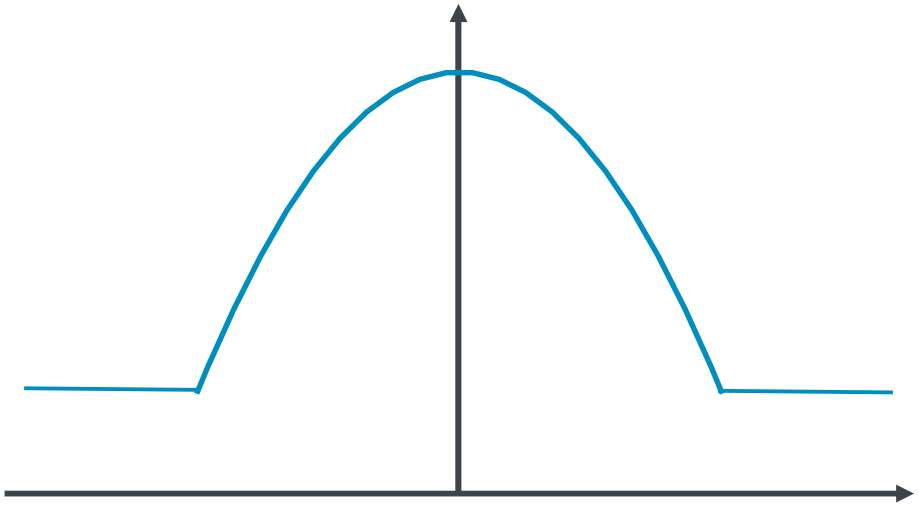
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$$\Delta t \approx \frac{N_1 \cdot L}{2 \cdot c \cdot n_1^2} NA^2$$





Graded-index fibers

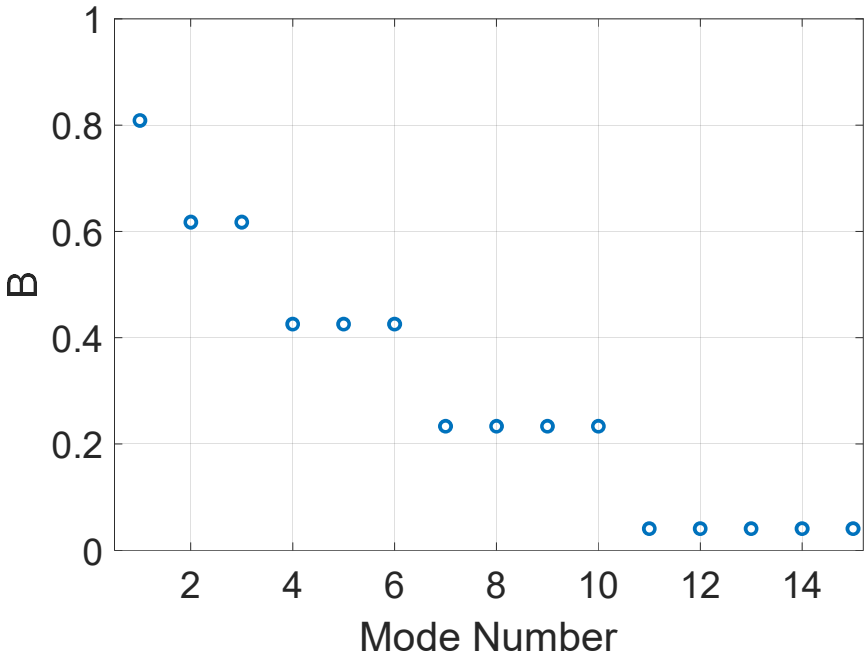


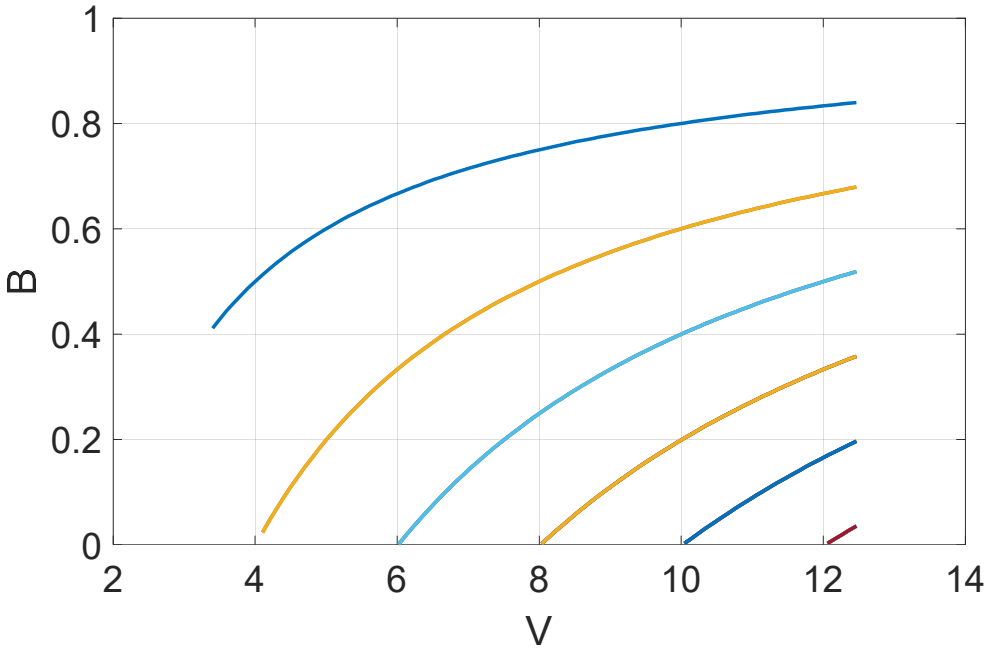
Graded-index fibers

Graded-index fibers

Graded-index fibers

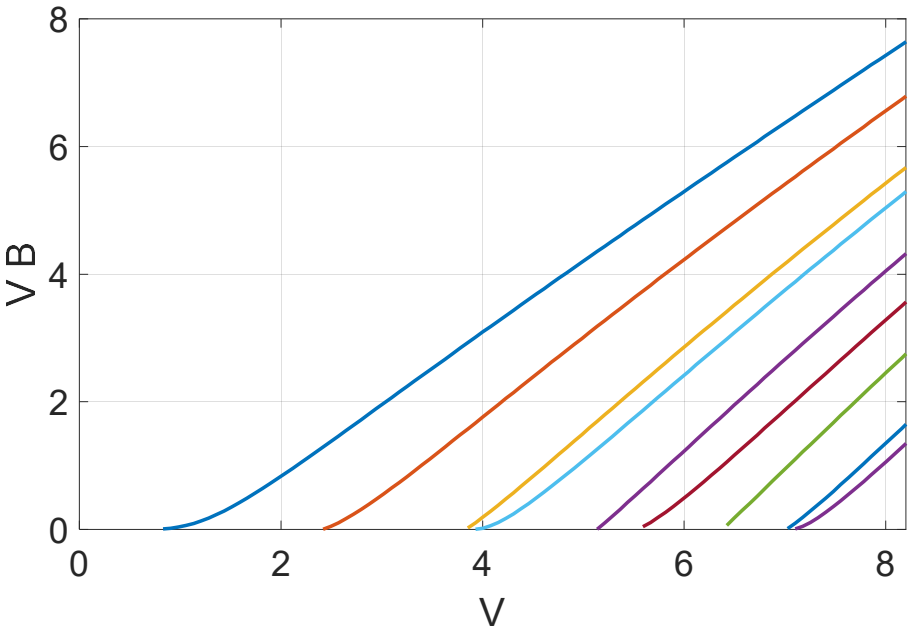
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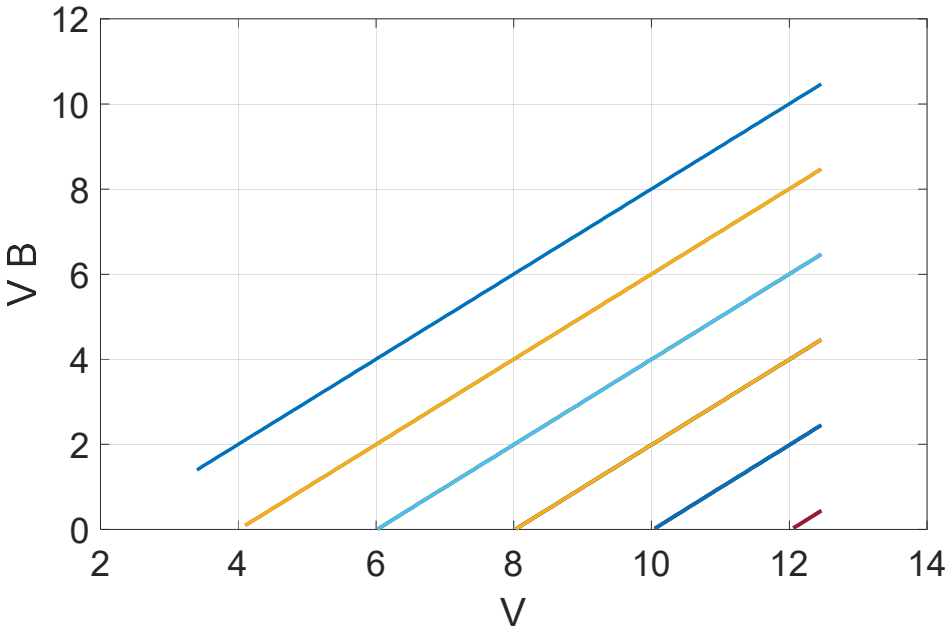




Graded-index fibers

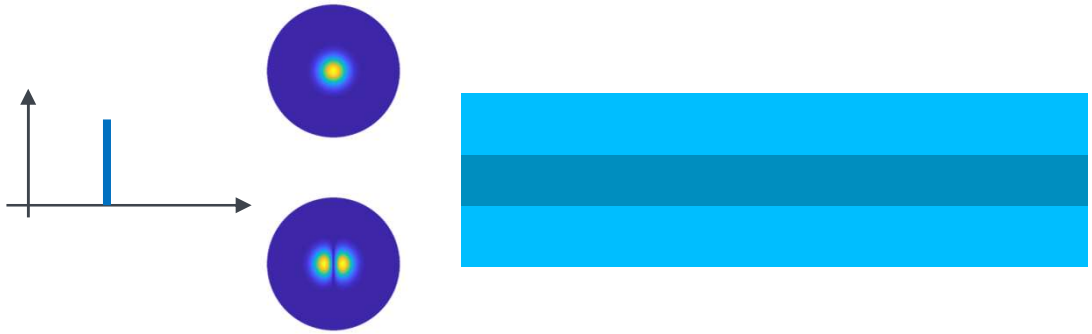
University of Stuttgart
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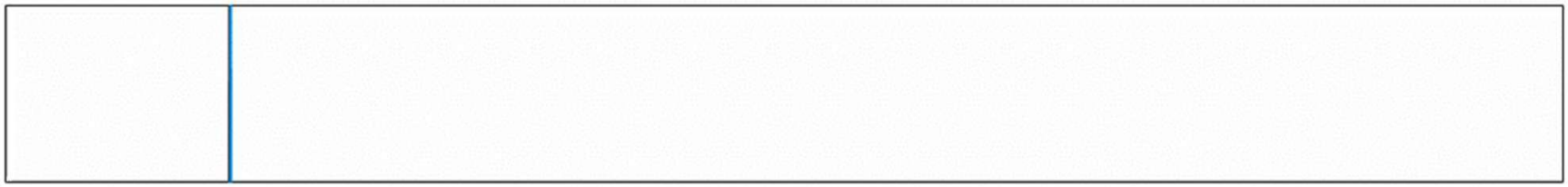




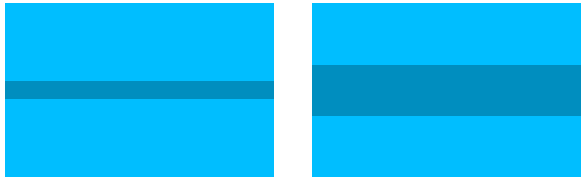
Graded-index fibers

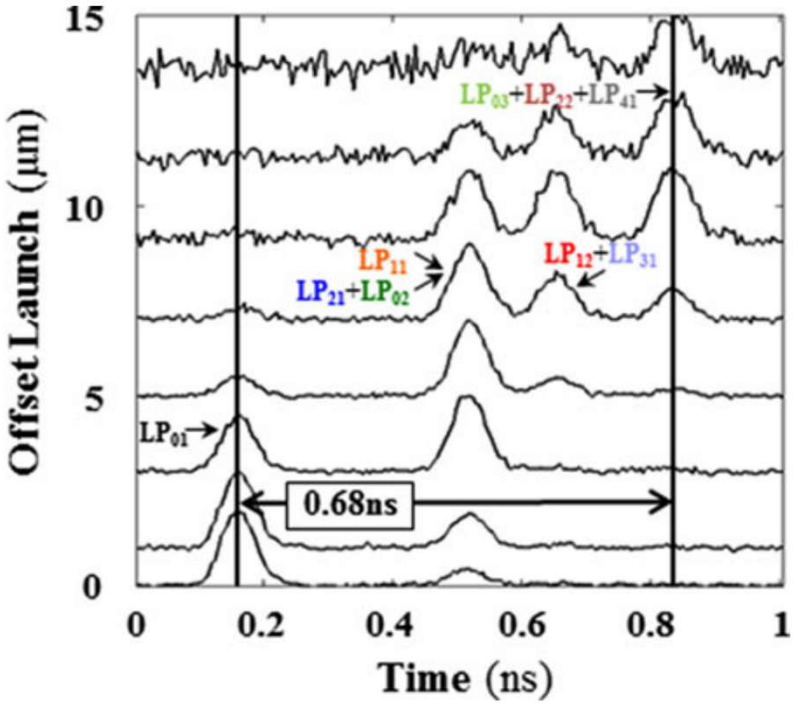
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Graded-index fibers

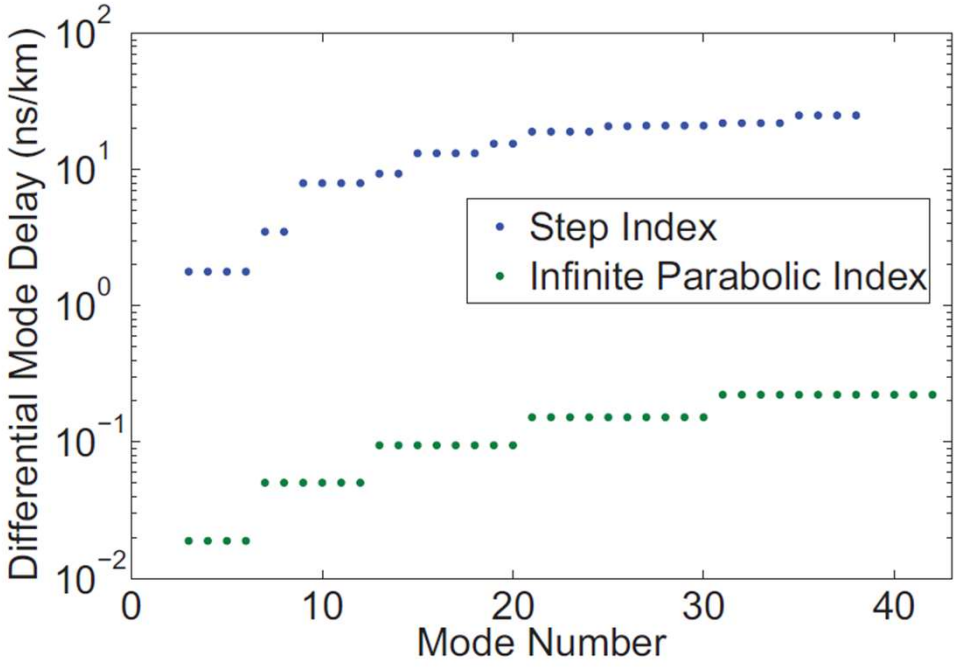




P. Sillard et al., Low-differential-mode-group-delay 9-LP-mode fiber, JLT 2016

Graded-index fibers

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$$\beta_{lp} = \cdot\left(2l + 4p - 2\right)^{1/2}$$



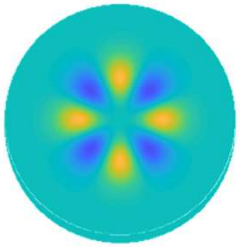
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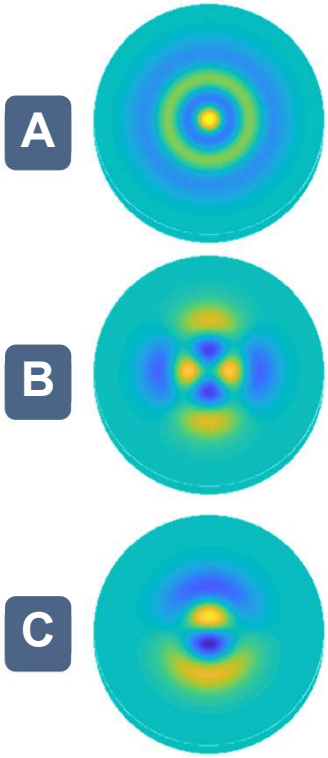
Concept Test

Concept test

University of Stuttgart
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Which mode is in the same group as the shown profile of LP41?





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University of Stuttgart
Institute of electrical and optical communications

Thank you!



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phone +49 (0) 711 6856 7901
www. int.uni-stuttgart.de/

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